

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: _____

Student ID: _____

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 only has S0/0/0 active (non-passive) for OSPF, which connects to R1. S0/0/1 is still passive, so R2 cannot form adjacency with R3. R3 reaches 192.168.2.0/24 via R1 instead.

R2 advertises routes to R1, but not directly to R3 due to the passive S0/0/1 interface.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

Cost = 65. R3 S0/0/1 cost (64) + R2 G0/0 cost (1) = 65. This is lower than via R1.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default reference bandwidth is 100 Mbps, so all links ≥ 100 Mbps get cost 1. This means Fast Ethernet, Gigabit Ethernet and 10G interfaces all have the same cost. Increasing the reference bandwidth allows OSPF to differentiate between these high-speed links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 cost (781) + R3 G0/0 cost (1) = 782.

192.168.23.0/30: R1 S0/0/1 cost (781) + R3 S0/0/0 cost (64) = 845,
or R1 S0/0/0 cost (781) + R2 S0/0/1 cost (64) = 845. Both paths have equal cost.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 845. Cost = R1 S0/0/1 (781) + R3 S0/0/0 (64) = 845 (or via R2 with same cost). After applying bandwidth 128 on all serial interfaces, costs remain the same because all serial links now have consistent bandwidth settings.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The ip ospf cost 1565 was applied to R1 S0/0/1, making the direct path to R3 cost = $1565 + 1 = 1566$. The path through R2 via S0/0/0 costs $781 + 781 + 1 = 1563$, which is lower. OSPF always selects the path with the lowest accumulated cost.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router ID uniquely identifies each router in the OSPF domain. It is used in DR/BDR elections and LSA origination. Without explicit control, the router ID depends on the highest active interface IP, which may change if an interface goes down. Using a stable, predictable router ID (via loopback or router-id command) ensures consistent OSPF operation.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR election only occurs on multi-access networks (e.g., Ethernet). The serial links between routers are point-to-point, so no DR/BDR is needed. The only broadcast network segments (G0/0 LANs) each have only one router, so DR/BDR election has no practical effect.

3. Why would you want to set an OSPF interface to passive?

Setting an interface to passive stops OSPF Hello packets from being sent on that interface. This reduces unnecessary OSPF traffic on LAN segments where no OSPF routers exist (e.g., end-user LANs), and also improves security by preventing unauthorized OSPF adjacencies.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of areas makes OSPF hierarchical. OSPF divides the network into areas (e.g., Area 0 as backbone). This reduces LSA flooding and SPF calculations, improving scalability.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be used. The routing decision is based on administrative distance (AD), not metric. OSPF AD=110, EIGRP AD=90, RIP AD=120. EIGRP has the lowest AD, so the EIGRP route wins.